

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 38

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• The nucleus of a stem cell from the endangered species is transferred into an empty egg of a closely related species (southern white rhino) • The embryo is implanted into a surrogate female of the related species which gives birth to a northern white rhino • Very low success rates with wildlife so far; deep-frozen cells could allow clones even after the last individuals die	3
01.2	• Micro-propagation: tissue culture producing many cell clusters from one plant, each grown into a genetically identical plant • Black poplar: cuttings / pollarded stems of pure native trees are rooted to produce many identical trees • Needed because the flowers are pollinated by introduced subspecies producing hybrids, so cloning maintains the pure gene pool (but no genetic diversity)	3
01.3	• Stores seeds of wild plants (especially threatened species and crop wild relatives) so species are not lost if extinct in the wild • Seeds collected worldwide; target of 10 000 seeds per species to maintain a diverse gene pool • Stored dry and refrigerated / frozen in an underground reinforced vault to reduce risk of accident • Seeds germinated periodically to check viability and replenish stocks; can be used for reintroduction / breeding	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 35-37; AQA spec 3.1.2.3.3.	

Question	Marking guidance	Marks
02.1	• Sigmoidal (S-shaped) growth curve • Slow start (few breeding individuals), then rapid / exponential growth (years 4-10) as abundant food and space mean birth rate far exceeds death rate • Growth slows as density-dependent factors (intra-species competition for food, disease) raise the death rate • Population stabilises / fluctuates around about 920-950 - the carrying capacity - where birth rate equals death rate	4
02.2	• Density-dependent: effect on survival increases with population density - food shortage / intra-species competition, disease spread, predation • Density-independent: chance of survival unaffected by density - drought, flood, volcanic eruption, severe winter	2
02.3	• Without wolves (natural predators removed) the population is regulated only by starvation / disease at a high density • At this density deer over-graze, preventing tree regeneration and damaging habitats / other species; culling keeps numbers below the damaging level (as in Scotland)	2
02.4	• Rising prey numbers give predators more food so predator numbers rise; high predation then reduces the prey population • Food shortage reduces predators, allowing prey to recover - cyclic fluctuations with predators lagging behind prey	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 78-81; AQA spec 3.1.3.4.	
03.1	• Surveys monitor photosynthesis rates, biomass, soil water content and the spread of pests / diseases so inputs are applied only where needed • Yield mapping of different parts of a field controls fertiliser application in each area for future crops • This maximises yield while avoiding wastage / reducing leaching and nutrient pollution	3
03.2	• Machinery, pesticides, fertilisers, improved breeds and infrastructure (transport, refrigeration) are expensive • They are mainly available in affluent societies; poorer farmers cannot afford the inputs	2
03.3	• Provides an income that allows producers to afford basic rights such as water, education, health care and food	1
03.4	• Nitrates are leached into water bodies used as sources of potable water • High nitrate levels cause blue baby syndrome (methaemoglobinaemia) and nitrates may be carcinogenic	2
03.5	<i>Any 2 from:</i> • Anaerobic digestion by microbes in rice padi fields • Livestock intestines (ruminant digestion) • Anaerobic decay of manure / slurry	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-332; AQA spec 3.5.1.3-3.5.1.4.	

Question	Marking guidance	Marks
04.1	• A: coppice rotation creates a patchwork of different ages / light levels - clearings with wildflowers for butterflies, dense regrowth for nesting birds; oak supports many invertebrates • B: same tree species but the closed canopy shades the floor so few flowering plants, fewer invertebrates and butterflies; fewer niches • C: non-indigenous single-species, single-age conifers support few native invertebrates; dense shade; no clearings or dead wood; needles acidify soil • Habitat diversity / structural variety increases the range of niches and hence species	5
04.2	<i>Any 3 from:</i> • Designated protected areas: SSSIs, NNRs (Sherwood Forest), SACs (Epping Forest), National Parks • Legal protection of ancient woodland: planning permission refused for developments destroying irreplaceable habitats (2012 NPPF) • Conservation management: coppicing, creating clearings, planting mixed native species, leaving dead wood, controlling deer • Planting new native woodlands linking fragments (corridors); Woodland Trust / National Trust ownership	3
04.3	• No species is completely dominant and food webs have many alternative links • So a change in the population of one species has a relatively small impact on the whole community	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	
05.1	• Extensive: low inputs of energy, food and chemicals so low cost and low environmental impact; sustainable; but low yield per area and dependent on natural productivity • Intensive: control of all limiting factors gives very high yield and profit per area; but high inputs (fishmeal from wild fish, energy), disease and lice at high density, pollution from wastes, antibiotic / pesticide use, escapes • Judgement depends on species, market and site	4
05.2	• Adults with desirable characteristics (fast growth, shape, colour) stripped of eggs and milt which are mixed for fertilisation • Eggs raised in aerated freshwater tanks, dead / diseased eggs removed daily; fry fed fishmeal pellets and moved through tanks as they grow • After 12-18 months smolts transferred to seawater cages, fed pellets until harvest at 3-5 kg	3
05.3	• About 10% of the world's mangroves have been cleared for coastal shrimp lagoons (up to 100 ha), removing nurseries and coastal protection • Lagoons need water exchange to remove wastes and aeration to prevent deoxygenation as wastes decompose • Eyestalk ablation stimulates breeding; high densities increase disease so antibiotics / pesticides used	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-356; AQA spec 3.5.2.3.	
06.1	• About 8 nights • The running mean fluctuates widely with few samples but stabilises at 21-22 from 8 nights onwards; extra samples do not change it significantly	2

Question	Marking guidance	Marks
06.2	• Moth activity varies with temperature, wind, rain and season (adults only fly for part of the year) so catches vary independently of population size • Sampling at different conditions / times would make results not comparable • Timing between samples should be chosen to detect the temporal variation of interest (eg weekly through the season)	3
06.3	• Accuracy: how close a recorded result is to the real value • Precision: the smallest interval between possible recorded values (eg a ruler marked in mm is more precise than one in cm)	2
06.4	• A null hypothesis (eg no difference / no correlation) can be tested statistically and accepted or rejected at a stated confidence level • Standardised techniques ensure results collected at different times, places or by different people are comparable • Other variables are controlled or monitored so their effect can be assessed; allows the study to be repeated / checked (reliability)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	
07.1	• Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis)	1
07.2	• Zebu tolerate heat but have a low milk yield; Ayrshire have a high milk yield but are not heat tolerant • Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield	2
07.3	<i>Any 2 from:</i> • Replacement of valuable animals that die with genetically identical individuals • Replacement of herds culled during a disease outbreak • Production of large numbers of individuals with desirable characteristics	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2.	
08.1	• Food: cod, tuna, swordfish, sharks, turtles; extinct - dodo, great auk, passenger pigeon • Fashion: leopard, snow leopard, tiger, ocelot fur; fur seals; crocodile / alligator leather; kingfisher, parrot, ostrich feathers • Entertainment / pets: parrots, lizards, snakes, tortoises, tropical marine fish (most wild-caught), dolphins / orcas for marine parks, wild-collected air plants • Furniture / ornaments: mahogany, teak, ramin timber; ebony and ivory piano keys; turtle shell, shark teeth, coral and shells as souvenirs • Traditional medicines: tiger (bones, claws), rhinoceros horn, seahorses	5
08.2	<i>Any 3 from:</i> • Threat to humans: sharks, venomous snakes, crocodiles • Disease vectors: malaria mosquitoes, tsetse flies • Predators of livestock: wolves, puma, birds of prey, herons, seals • Agricultural / forestry pests and competitors: rabbits, deer, wood-boring beetles, squirrels, beavers	3

Question	Marking guidance	Marks
08.3	• High demand and prices make illegal trade / poaching profitable; laws are hard to enforce across borders • Belief in the medicine persists despite little scientific evidence; species become rarer and more valuable	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-21; AQA spec 3.1.2.2.	
09.1	• All greenhouse gases expressed as CO₂ equivalent (allowing for different warming strengths) • Direct emissions from fuel combustion plus indirect: manufacture of components, maintenance items, final disposal (life-cycle analysis) • A fair share of the infrastructure: roads, car parks, signals, street lighting	3
09.2	• Compares the populations of over 18 000 populations of 3700 vertebrate species (terrestrial, freshwater, marine) with 1970 levels • Published in the Living Planet Report as a measure of the state of the planet's ecosystems / biodiversity loss over time	2
09.3	• Hardy, disease-resistant Boran cattle; regular tick dipping reduces ticks for wild herbivores too • Mobile bomas (night enclosures) protect cattle from lions and leave dung that enriches soil and plant growth • Electric fences keep game in and poachers out; biological corridors let animals (except rhinos) move to other reserves; problem elephants relocated • Monitoring: radio-collared lions, drones, camera traps, track surveys; endangered species (Grevy's zebra, northern white rhino) protected • Community links: ecotourism income funds health, education and water projects; local employment; cattle breeding help for local farmers	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 380-383; AQA spec 3.6.4.2.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Resources: wood, fibres (cotton, wool, silk), oils, fuels (wood, biofuels), food; new food species (morama bean, perennial cereals, eland, bison) better adapted to local conditions • Biomimetics: vehicle design, architecture, structures, adhesion, materials, ultrasound • Medicines from plants and animals (examples); most species unstudied - argument for habitat conservation • Physiological research using wildlife species; wildlife as pest control agents (ladybirds, Encarsia, Cactoblastis, Bt) • Genetic resources: crop wild relatives, Vavilov centres, gene pools • Ecosystem services: atmospheric composition, hydrological and biogeochemical cycles, soil maintenance, pollination, seed dispersal, habitat provision • Ethical / aesthetic / recreational arguments; the unknown value of species not yet investigated (precautionary principle) • Evaluation: which arguments are most persuasive to decision makers; economic valuation vs intrinsic value; judgement	9
10.2	• The variety and abundance of life (species, genes, habitats) in an area / on Earth	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Question	Marking guidance	Marks
11.1	Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Sustainability definitions; population and per capita impact; attitudes of rural and urban societies • Negative feedback: cloud cover and albedo, CO₂ and photosynthesis, evaporation and precipitation, density-dependent population regulation • Positive feedback driven by human activities: permafrost, acidification, albedo decline, methane hydrates, fires, cirrus clouds, decomposition • Tipping points: rare under negative feedback; large events (Permian methane hydrate); self-sustaining change from human activities • Diversity and resistance: reefs and rainforests vs low-diversity agroecosystems lacking detritivores, pollinators and predators • Human responses that override feedback (fertilisers, irrigation, pesticides); links to climate change chapter; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	25

Question	Marking guidance	Marks
11.2	Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Temperate broadleaf woodland: seasonality, deep soils, high biodiversity, resources, climate and soil regulation; deforestation, fragmentation, management change; SSSIs, ancient woodland protection, coppicing, mixed planting • Tropical rainforest: stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils; fuelwood, timber, agriculture, mining, reservoirs, climate change; protected areas, debt-for-nature swaps, sustainable exploitation • Tropical coral reefs: cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity; fisheries, erosion protection, medicines, tourism; damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species; MPAs, no-take zones, control of activities • Deep-water reefs: cold, dark, slow growth; research, fisheries; trawling, oil / gas, acidification; SACs • Oceanic islands: isolation, endemism, no mammal predators; exploitation, introduced species, habitat change, sea level; eradication, control of development and visitors • Mangroves: halophytes, anaerobic mud; erosion protection, fisheries nurseries, sediment trapping; clearance for aquaculture / urban / ports, pollution, climate change; protection, replanting • Antarctica: extreme cold, seasonality, marine nutrients, ice; water store, albedo, carbon, research; climate change, ozone, tourism, fishing, minerals; Antarctic Treaty, CCAMLR, tourism and waste rules • Synthesis: features -> importance -> threats -> strategy; evaluation of success; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	25